

HOMework 3

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Instructions: Although this is a programming homework, you only need to hand in a pdf answer file. There is no need to submit the latex source or any code. You can choose any programming language, as long as you implement the algorithm from scratch.

Use this latex file as a template to develop your homework. Submit your homework on time as a single pdf file to **Gradescope**. Please check Piazza for updates about the homework.

1 Questions (50 pts)

1. (9 pts) Explain whether each scenario is a classification or regression problem. And, provide the number of data points (n) and the number of features (p).

- (a) (3 pts) We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in predicting CEO salary with given factors.

This is a regression problem because the target variable, CEO salary, is a continuous quantity.

The number of data points is given by:

$$n = 500$$

The number of features is given by:

$$p = 3$$

since the factors used to predict the CEO salary are profit, number of employees, and industry.

- (b) (3 pts) We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

This is a classification problem because the target variable, whether the product is a success or failure, is categorical.

The number of data points is given by:

$$n = 20$$

The number of features is given by:

$$p = 13$$

since we have recorded price charged for the product, marketing budget, competition price, and ten other variables.

- (c) (3 pts) We are interesting in predicting the % change in the US dollar in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the dollar, the % change in the US market, the % change in the British market, and the % change in the German market.

This is a regression problem because the target variable, the percentage change in the US dollar, is a continuous quantity.

The number of data points is given by:

$$n = 52$$

since we collect weekly data for the year 2012.

The number of features is given by:

$$p = 3$$

as we record the percentage changes in the US, British, and German markets.

2. (6 pts) The table below provides a training data set containing six observations, three predictors, and one qualitative response variable.

X_1	X_2	X_3	Y
0	3	0	Red
2	0	0	Red
0	1	3	Red
0	1	2	Green
-1	0	1	Green
1	1	1	Red

Suppose we wish to use this data set to make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K-nearest neighbors.

- (a) (2 pts) Compute the Euclidean distance between each observation and the test point, $X_1 = X_2 = X_3 = 0$.

- i. $d = \sqrt{0^2 + 3^2 + 0^2} = 3$
- ii. $d = \sqrt{2^2 + 0^2 + 0^2} = 2$
- iii. $d = \sqrt{0^2 + 1^2 + 3^2} = \sqrt{10}$
- iv. $d = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5}$
- v. $d = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$
- vi. $d = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

- (b) (2 pts) What is our prediction with $K = 1$? Why?

The closest neighbour is the fifth one, with a distance of $\sqrt{2}$, and its label is green. Hence, the prediction is **green**:

- (c) (2 pts) What is our prediction with $K = 3$? Why?

The three nearest neighbors' labels are:

- Green
- Red
- Red

Since the majority class is Red (2 out of 3), the prediction is **red**:

3. (12 pts) When the number of features p is large, there tends to be a deterioration in the performance of KNN and other local approaches that perform prediction using only observations that are near the test observation for which a prediction must be made. This phenomenon is known as the curse of dimensionality, and it ties into the fact that non-parametric approaches often perform poorly when p is large.

- (a) (2pts) Suppose that we have a set of observations, each with measurements on $p = 1$ feature, X . We assume that X is uniformly (evenly) distributed on $[0, 1]$. Associated with each observation is a response value. Suppose that we wish to predict a test observation's response using only observations that are within 10% of the range of X closest to that test observation. For instance, in order to predict the response for a test observation with $X = 0.6$, we will use observations in the range $[0.55, 0.65]$. On average, what fraction of the available observations will we use to make the prediction?

Since X is uniformly distributed, the fraction of the available observations used is proportional to the length of the interval.

Thus, the fraction of observations is:

$$\frac{0.1}{1} = 0.1$$

- (b) (2pts) Now suppose that we have a set of observations, each with measurements on $p = 2$ features, X_1 and X_2 . We assume that predict a test observation's response using only observations that (X_1, X_2) are uniformly distributed on $[0, 1] \times [0, 1]$. We wish to be within 10% of the range of X_1 and within 10% of the range of X_2 closest to that test observation. For instance, in order to predict the response for a test observation with $X_1 = 0.6$ and $X_2 = 0.35$, we will use observations in the range $[0.55, 0.65]$ for X_1 and in the range $[0.3, 0.4]$ for X_2 . On average, what fraction of the available observations will we use to make the prediction?

The fraction of observations within this range for X_1 is:

$$\frac{0.1}{1} = 0.1$$

The fraction of observations within this range for X_2 is:

$$\frac{0.1}{1} = 0.1$$

Since X_1 and X_2 are independent, the total fraction of observations used is:

$$0.1 \times 0.1 = (0.1)^2$$

- (c) (2pts) Now suppose that we have a set of observations on $p = 100$ features. Again the observations are uniformly distributed on each feature, and again each feature ranges in value from 0 to 1. We wish to predict a test observation's response using observations within the 10% of each feature's range that is closest to that test observation. What fraction of the available observations will we use to make the prediction?

Following the proof in the previous parts, since each axis is independent, the total fraction of observations is:

$$(0.1)^p$$

therefore for $p = 100$, the total fraction of observations is:

$$(0.1)^{100}$$

- (d) (3pts) Using your answers to parts (a)–(c), argue that a drawback of KNN when p is large is that there are very few training observations “near” any given test observation.

As the dimensionality p increases, the fraction of observations near the test point decreases rapidly. In general, for p features, the fraction of observations near the test point is given by:

$$0.1^p$$

which rapidly decreases as p increases.

This illustrates as p grows larger, the space becomes sparse, and very few training observations are close to any given test observation. As a result, KNN performs poorly in high dimensions because it becomes difficult to find enough nearby neighbors for accurate prediction.

- (e) (3pts) Now suppose that we wish to make a prediction for a test observation by creating a p -dimensional hypercube centered around the test observation that contains, on average, 10% of the training observations. For $p = 1, 2$, and 100, what is the length of each side of the hypercube? Comment on your answer.

The volume of the hypercube is given by L^p , where L is the length of each side of the hypercube. We want:

$$L^p = 0.1$$

$$L = 0.1^{1/p}$$

For $p = 1$, length is:

$$L = 0.1$$

For $p = 2$, length is:

$$L = 0.316$$

For $p = 100$, length is:

$$L = 0.997$$

As p increases, the length of each side approaches 1. This shows that in high-dimensional spaces, to capture 10% of the training data, the neighborhood must expand significantly, eventually including almost the entire space.

4. (6 pts) Suppose you trained a classifier for a spam detection system. The prediction result on the test set is summarized in the following table.

		Predicted class	
		Spam	not Spam
Actual class	Spam	8	2
	not Spam	16	974

Calculate

- (a) (2 pts) Accuracy

$$Accuracy = \frac{tp + tn}{tp + fn + fp + tn}$$

Therefore,

$$Accuracy = \frac{982}{1000}$$

- (b) (2 pts) Precision

$$Precision = \frac{tp}{tp + fp}$$

Therefore,

$$Precision = \frac{8}{24} = \frac{1}{3}$$

- (c) (2 pts) Recall

$$Recall = \frac{tp}{tp + fn}$$

Therefore,

$$Recall = \frac{8}{10} = \frac{4}{5}$$

5. (9pts) Again, suppose you trained a classifier for a spam filter. The prediction result on the test set is summarized in the following table. Here, "+" represents spam, and "-" means not spam.

Confidence positive	Correct class
0.95	+
0.85	+
0.8	-
0.7	+
0.55	+
0.45	-
0.4	+
0.3	+
0.2	-
0.1	-

(a) (6pts) Draw a ROC curve based on the above table.

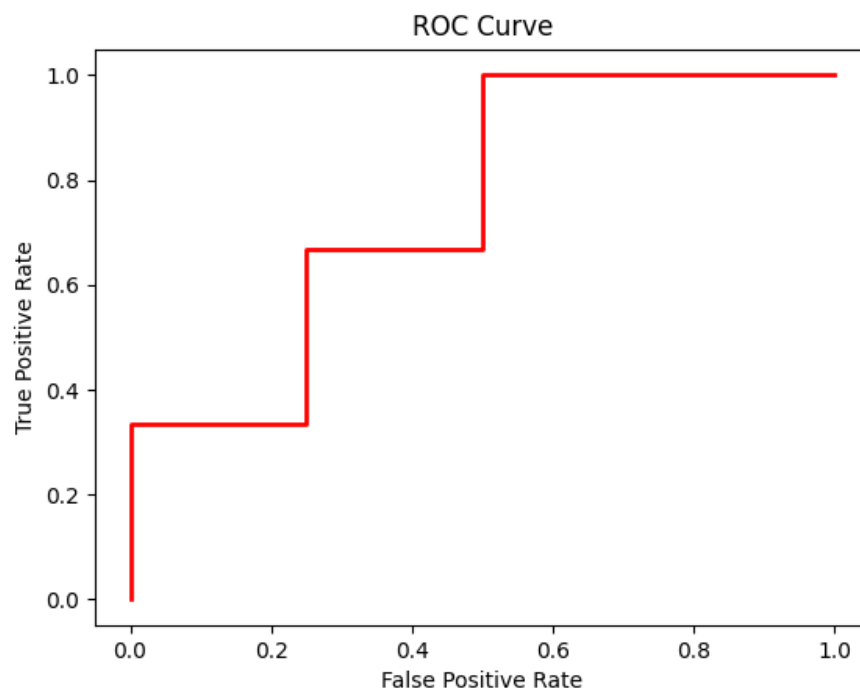


Figure 1: ROC Curve

(b) (3pts) (Real-world open question) Suppose you want to choose a threshold parameter so that mails with confidence positives above the threshold can be classified as spam. Which value will you choose? Justify your answer based on the ROC curve.

Let's try all possible threshold ranges.

i. $t \geq 0.95$:

- TP: 1
- FN: 5
- TN: 4
- FP: 0

$$Accuracy : \frac{5}{10} = 0.5$$

$$Precision : 1$$

$$\text{Recall} : \frac{1}{6} \approx 0.167$$

ii. $0.95 > t \geq 0.85$:

- TP: 3
- FN: 4
- TN: 4
- FP: 0

$$\text{Accuracy} : \frac{7}{10} = 0.7$$

$$\text{Precision} : 1$$

$$\text{Recall} : \frac{1}{3} \approx 0.334$$

iii. $0.85 > t \geq 0.8$:

- TP: 2
- FN: 4
- TN: 3
- FP: 1

$$\text{Accuracy} : \frac{5}{10} = 0.5$$

$$\text{Precision} : \frac{2}{3} = 0.667$$

$$\text{Recall} : \frac{2}{6} \approx 0.334$$

iv. $0.8 > t \geq 0.7$:

- TP: 3
- FN: 3
- TN: 3
- FP: 1

$$\text{Accuracy} : \frac{6}{10} = 0.6$$

$$\text{Precision} : \frac{3}{4} = 0.75$$

$$\text{Recall} : \frac{3}{6} = 0.5$$

v. $0.7 > t \geq 0.55$:

- TP: 4
- FN: 2
- TN: 3
- FP: 1

$$\text{Accuracy} : \frac{7}{10} = 0.7$$

$$\text{Precision} : \frac{4}{5} = 0.8$$

$$\text{Recall} : \frac{4}{6} \approx 0.667$$

vi. $0.55 > t \geq 0.45$:

- TP: 4
- FN: 2
- TN: 2

- FP: 2

$$Accuracy : \frac{6}{10} = 0.6$$

$$Precision : \frac{4}{6} \approx 0.667$$

$$Recall : \frac{4}{6} \approx 0.667$$

vii. $0.45 > t \geq 0.4$:

- TP: 5
- FN: 1
- TN: 2
- FP: 2

$$Accuracy : \frac{7}{10} = 0.7$$

$$Precision : \frac{5}{7} \approx 0.714$$

$$Recall : \frac{5}{6} \approx 0.8334$$

viii. $0.4 > t \geq 0.3$:

- TP: 6
- FN: 0
- TN: 2
- FP: 2

$$Accuracy : \frac{8}{10} = 0.8$$

$$Precision : \frac{6}{8} = 0.75$$

$$Recall : \frac{6}{6} = 1$$

ix. $0.3 > t \geq 0.2$:

- TP: 6
- FN: 0
- TN: 1
- FP: 3

$$Accuracy : \frac{7}{10} = 0.7$$

$$Precision : \frac{6}{9} \approx 0.67$$

$$Recall : \frac{6}{6} = 1$$

x. $0.2 > t \geq 0.1$:

- TP: 6
- FN: 0
- TN: 0
- FP: 4

$$Accuracy : \frac{6}{10} = 0.6$$

$$Precision : \frac{6}{10} = 0.6$$

$$Recall : \frac{6}{6} = 1$$

Looking at the accuracy, precision values, for a problem of spam detection, accuracy and precision should be high. So, a **threshold t of $0.4 > t \geq 0.3$ on the confidence positive would be good.**

Also, if we observe the ROC graph, we can see that this threshold is correct as:

- At this threshold, we have achieved high TPR (also called recall) of 1 because we are capturing all spam emails.
- The precision (0.75) is slightly lower, which indicates that 25% of emails predicted as spam are actually not spam. However, a precision of 0.75 is still relatively high, which means the classifier is effective at identifying spam without generating too many false positives.
- The accuracy (80%) at this threshold is also the highest compared to other thresholds.

6. (8 pts) In this problem, we will walk through a single step of the gradient descent algorithm for logistic regression. Assume two dimension input. Recap:

$$f(x; w, b) = \sigma(w \cdot x + b)$$

$$\text{Cross entropy loss } L(\hat{y}, y) = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

$$\text{The single update step } \theta^{t+1} = \theta^t - \eta \nabla_{\theta} L(f(x; \theta), y), \text{ where } \theta = [w_1, w_2, b]^T$$

Now given

$$\text{Initial parameters : } w_1 = w_2 = b = 0, (\Rightarrow \theta^0 = [0, 0, 0])$$

$$\text{Learning rate } \eta = 0.1$$

$$\text{data example : } x = [3, 2], y = 1$$

- (a) (4 pts) Compute the first gradient $\nabla_{\theta} L(f(x; \theta), y)$.

First, we compute $f(x; \theta)$:

$$f(x; \theta) = \sigma(0 \cdot 3 + 0 \cdot 2 + 0) = \sigma(0) = \frac{1}{2}$$

Now, we compute the gradient of the loss function:

$$\nabla_{\theta} L(f(x; \theta), y) = \nabla_{f(x; \theta)} L(f(x; \theta), y) \cdot \nabla_{w \cdot x + b} f(x; \theta) \cdot \nabla_{\theta} w \cdot x + b$$

$$\nabla_{\theta} L(f(x; \theta), y) = \frac{(f(x; \theta) - y)}{f(x; \theta)(1 - f(x; \theta))} \cdot f(x; \theta)(1 - f(x; \theta)) \cdot \nabla_{\theta} (w_1 x_1 + w_2 x_2 + b)$$

Also for $\nabla_{\theta} (w_1 x_1 + w_2 x_2 + b)$,

$$\frac{\partial (w_1 x_1 + w_2 x_2 + b)}{\partial w_1} = x_1$$

$$\frac{\partial (w_1 x_1 + w_2 x_2 + b)}{\partial w_2} = x_2$$

$$\frac{\partial (w_1 x_1 + w_2 x_2 + b)}{\partial b} = 1$$

Therefore,

$$\nabla_{\theta} (w \cdot x + b) = \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

Therefore,

$$\nabla_{\theta} L(f(x; \theta), y) = (f(x; \theta) - y) \cdot \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

Therefore,

$$\nabla_{\theta} L(f(x; \theta), y) = \left(\frac{1}{2} - 1 \right) \cdot \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = -\frac{1}{2} \cdot \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} \\ -1 \\ -\frac{1}{2} \end{bmatrix}$$

- (b) (4 pts) Compute the updated parameter vector θ^1 from the single update step.

$$\theta^1 = \theta^0 - \eta \nabla_{\theta} L(f(x; \theta), y)$$

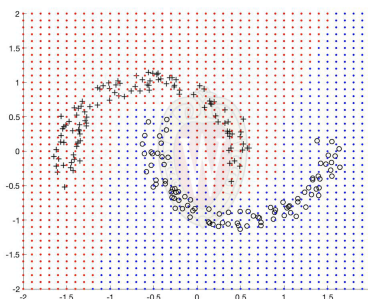
Therefore,

$$\begin{aligned} \theta^1 &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} -\frac{3}{2} \\ -1 \\ -\frac{1}{2} \end{bmatrix} \\ \theta^1 &= \begin{bmatrix} 0.15 \\ 0.1 \\ 0.05 \end{bmatrix} \end{aligned}$$

2 Programming (50 pts)

- (10 pts) Use the whole D2z.txt as training set. Use Euclidean distance (i.e. $A = I$). Visualize the predictions of 1NN on a 2D grid $[-2 : 0.1 : 2]^2$. That is, you should produce test points whose first feature goes over $-2, -1.9, -1.8, \dots, 1.9, 2$, so does the second feature independent of the first feature. You should overlay the training set in the plot, just make sure we can tell which points are training, which are grid.

The expected figure looks like this.



The generated figure is given below:

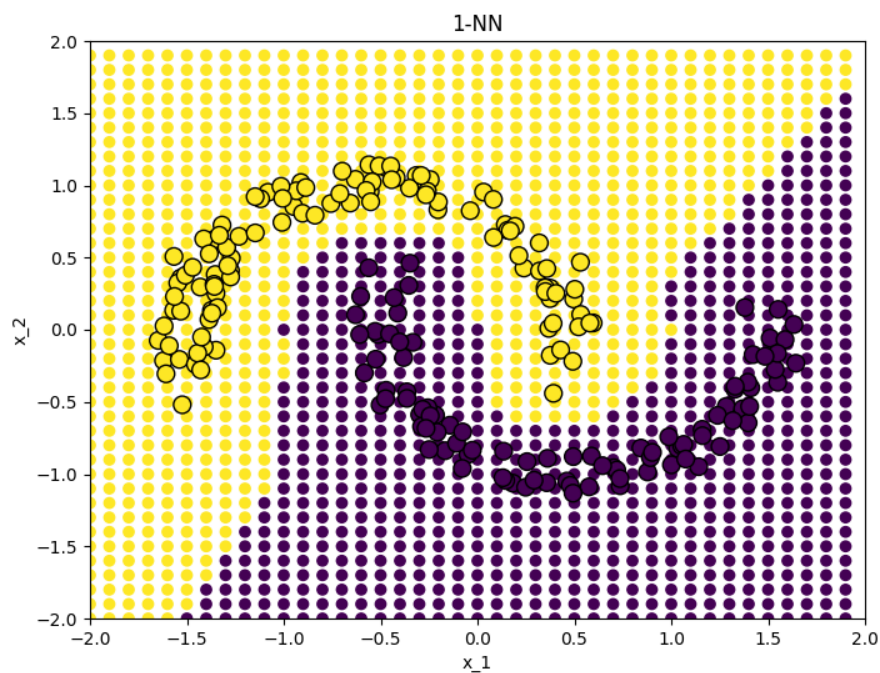


Figure 2: 1 NN

Spam filter Now, we will use 'emails.csv' as our dataset. The description is as follows.

	Features																			Label	
	the	to	ect	and	for	of	a	you	hou	in	...	connevey	jay	valued	lay	infrastructure	military	allowing	ff	dry	Prediction
Email No.																					
Email 1	0	0	1	0	0	0	2	0	0	0	...	0	0	0	0		0	0	0	0	0
Email 2	8	13	24	6	6	2	102	1	27	18	...	0	0	0	0		0	0	0	1	0
Email 3	0	0	1	0	0	0	8	0	0	4	...	0	0	0	0		0	0	0	0	0
Email 4	0	5	22	0	5	1	51	2	10	1	...	0	0	0	0		0	0	0	0	0
Email 5	7	6	17	1	5	2	57	0	9	3	...	0	0	0	0		0	0	0	1	0

- Task: spam detection
- The number of rows: 5000
- The number of features: 3000 (Word frequency in each email)
- The label (y) column name: 'Predictor'
- For a single training/test set split, use Email 1-4000 as the training set, Email 4001-5000 as the test set.
- For 5-fold cross validation, split dataset in the following way.
 - Fold 1, test set: Email 1-1000, training set: the rest (Email 1001-5000)

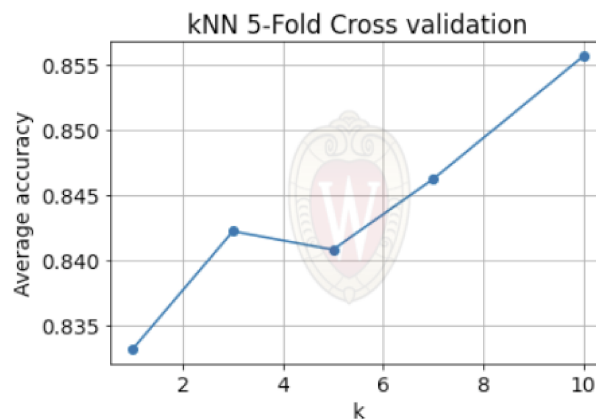
- Fold 2, test set: Email 1000-2000, training set: the rest
 - Fold 3, test set: Email 2000-3000, training set: the rest
 - Fold 4, test set: Email 3000-4000, training set: the rest
 - Fold 5, test set: Email 4000-5000, training set: the rest
2. (10 pts) Implement 1NN, Run 5-fold cross validation. Report accuracy, precision, and recall in each fold.

Fold	Accuracy	Precision	Recall
1	0.825	0.653631	0.821053
2	0.853	0.685714	0.866426
3	0.860	0.718182	0.834507
4	0.851	0.716418	0.816327
5	0.775	0.605744	0.758170

3. (10 pts) Implement logistic regression, Run 5-fold cross validation. Report accuracy, precision, and recall in each fold.

Fold	Accuracy	Precision	Recall
1	0.971	0.926667	0.975439
2	0.973	0.940141	0.963899
3	0.967	0.934256	0.950704
4	0.956	0.928082	0.921769
5	0.941	0.889590	0.921569

4. (10 pts) Run 5-fold cross validation with kNN varying k (k=1, 3, 5, 7, 10). Plot the average accuracy versus k, and list the average accuracy of each case.
Expected figure looks like this.



The generated plot looks like this:

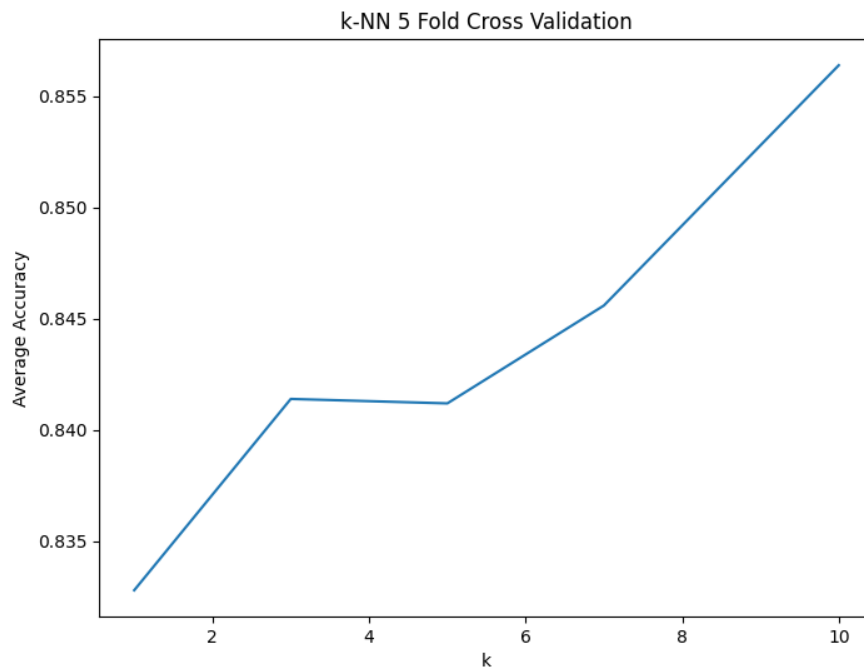
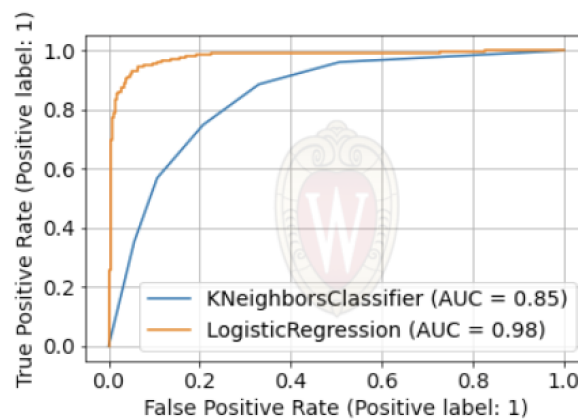


Figure 3: Knn 5 Fold

5. (10 pts) Use a single training/test setting. Train kNN (k=5) and logistic regression on the training set, and draw ROC curves based on the test set. Expected figure looks like this.



Note that the logistic regression results may differ.

The generated plot is as follows:

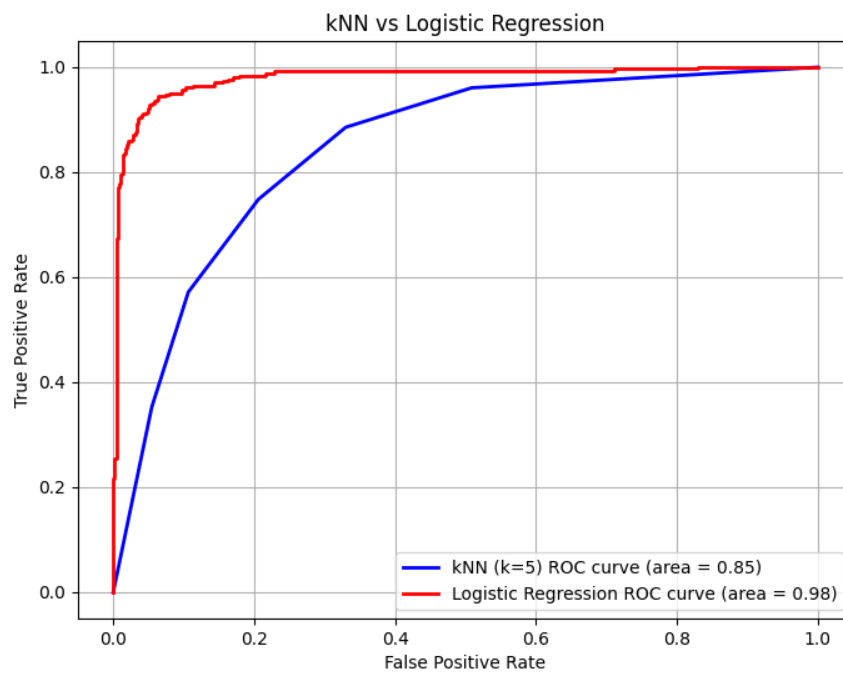


Figure 4: Knn vs Logistic ROC